SSQ-LR: Sub-FP16 Recurrent State Quantization for Hybrid Reasoners

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Abstract

This is a living paper shell for a preregistered Mac-first experiment. SSQ-LR tests whether recurrent SSM state in hybrid reasoning models can be quantized below FP16 while preserving quality. No positive result or native systems claim is made yet.

1 Current Gate

The branch must pass state-distribution heterogeneity, state-only quantization sensitivity, and cross-model transfer gates before any GPU validation. S1 must first show real state heterogeneity on a live hybrid model; S2 then freezes one state-only quantization recipe; S3 tests that recipe without per-model retuning. Failing any gate kills the branch before GPU work. The current real trace packet builder/checker covers S1. S2 and S3 now have executable follow-up contracts in `experimental/shared/followup_gate_contracts.py`: they require frozen recipe IDs, byte accounting, BF16 no-op controls, paired confidence intervals, frozen recipe/source hashes, and no-retuning transfer rows. These contracts are not model evidence: no quantization-quality, byte-savings, or cross-model-transfer claim is allowed until a real S1 packet promotes and the S2/S3 contracts are filled from the frozen S1 trace/replay surface.

2 Mac Artifact Status

A deterministic synthetic S1 packet now validates the real-schema checker path before real hybrid-model state dumps. It emits 288 rows covering 12 prompts, 6 recurrent layers, and the four preregistered buckets, producing `SCHEMA_REHEARSAL_NOT_PROMOTABLE_SYNTHETIC_SSQ_LR_S1`. This is synthetic-only: it is not model evidence, not a quality result, and not GPU evidence. The metadata-only model eligibility packet identifies live hybrid targets (`granite-4.0-h-tiny`, `granite-4.0-h-small`, `granite-4.0-h-small-FP8`, and `Qwen3-Next-80B-A3B-Instruct`) and ties them to shared architecture-map hashes, but all remain blocked for real S1 on this Mac because no corresponding model weights are cached repo-locally.

3 Next Evidence

The next admissible row is a real SSM-state dump from the smallest available hybrid reasoner. Promotion requires S1 heterogeneity on real states before any state-quantization recipe is tuned.

4 Admissible Real Packet

The real S1 packet must include model and tokenizer revisions, prompt manifest hash, `prompt_ids_hash`, seed list, context lengths, dtype, device, command, and the `architecture_map_hash` from the shared config-derived maps plus the `trace_plan_hash` from the frozen SSQ-LR trace plan. Promotable packets must also include the exact `trace_plan_path`; its file SHA-256 must match the registered `trace_plan_hash`, so caller-created subsets are resource-limited readouts rather than S1 promotion evidence. The current capture-manifest templates are generated by `experimental.shared.hybrid_trace_capture_manifest`; they must be filled from real SSM-state captures before packet building. Served HF IDs are preserved as `served_model_id` but rows are validated against the canonical architecture-map `model_id`. Rows must report `model_id`, `model_revision`, `prompt_id`, `layer`, `position_bucket`, `state_shape`, `max_abs`, `rms`, `std`, `kurtosis`, `outlier_mass`, and a `bf16_no_quant` control. The packet is valid only if every prompt/layer pair covers `prefill_end`, `2k_or_end`, `8k_or_end`, and `final_minus_128` position buckets and at least 12 prompt IDs unless `config.json` records an explicit resource-limit note. Rows must identify an SSM/Mamba `layer_kind` and recurrent `state_tensor_kind`, so arbitrary activation tensors cannot promote S1. Any resource-limited packet must use a `RESOURCE_LIMITED_NOT_PROMOTABLE` decision. A schema rehearsal must set `schema_rehearsal: true` and use a `SCHEMA_REHEARSAL_NOT_PROMOTABLE` decision. The summary must include the recomputed S1 evaluator fields: gate name/status/pass flag, prompt count, position buckets, SSM layer count, required and observed passing-layer counts, distribution-passing layer counts, selected S1 ratio and prompt-level lower bound, Holm-adjusted two-sample readout, magnitude/distribution pass flags, a 1.25x distribution-effect floor flag, and final-minus-128 versus prefill-end max-abs/std/kurtosis ratios. The checker recomputes these fields from raw rows rather than global layer means and rejects stale summaries; a tiny distribution-only shift cannot promote S1, and a global mean spike in one layer cannot satisfy the distribution path unless the counted layer/metric tests also clear the per-layer 1.25x effect-size floor. It must pass `experimental.shared.check_gate_packet --mode real --project ssq_lr`. Saved tensor packets must be converted with `experimental.shared.hybrid_trace_packet_builder` before validation. The converted packet must include `tensors/tensor_manifest.json` and the referenced tensor artifacts; each raw row copies `tensor_name`, `tensor_source_name`, `tensor_storage_name`, `tensor_sha256`, `tensor_dtype`, and `tensor_shape` from that manifest, and `state_shape` must match the saved `tensor_shape`.

Required baseline	Purpose
BF16 no-op replay	Catch hook/serialization bugs.
INT8 state-only	Standard low-bit state control.
FP8-style state-only	Checks whether sub-FP16 state can match the likely deployed fallback.
MXFP4 state-only	Candidate sub-FP16 recipe.
Random same-L2 noise	Separate structure from noise magnitude.
Scale shuffle	Test scale assignment sensitivity.
Byte accounting	Include scale/metadata overhead.

Table 1: Controls required before SSQ-LR can claim a state-quantization recipe.

5 Reviewer Packet

The reviewer-facing claim boundary is tracked in `experimental/ssq_lr/paper/reviewer_pack.md`. It records that SSQ-LR is not camera-ready as a method paper until real S1–S3 evidence exists.

6 Claim Boundary

SSQ-LR is only about recurrent SSM-state precision in hybrid reasoners. It does not claim weight quantization, activation quantization, KV-cache quantization, native serving speed, or memory savings until a real state recipe and later GPU validation pass their preregistered gates.